Notes: Dinc and Gupta (JF, 2011)

The Decision to Privatize: Finance and Politics

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1 Big picture

- **Question.** Which government-owned firms are selected for privatization, and how much do financial characteristics, electoral incentives, and political patronage affect that decision?
- **Setting.** The paper studies federal government-owned nonfinancial firms in India after the start of economic reforms in 1991.
- **Why this setting is useful.** India is a competitive multiparty democracy, and the authors observe both firms that were privatized and firms that remained fully government-owned. This makes it possible to study selection into privatization rather than only outcomes after privatization.
- **Core friction.** Privatization has broad social benefits, such as sale proceeds, efficiency improvements, financial-market development, and fiscal relief. But the costs are concentrated: local workers may lose rents or jobs, and politicians may lose private benefits from controlling state-owned firms.
- **Main financial result.** Larger firms are privatized earlier, while firms with larger wage bills are privatized later.
- **Main political result.** Privatization is delayed when a firm is located in a constituency where the governing and opposition party alliances are in a close electoral race.
- **Opposition-strength result.** Privatization is also delayed where the opposition party alliance has more local voter support.
- **Patronage result.** No firm located in the home state of the cabinet minister with jurisdiction over that firm is privatized.
- **Performance result.** Using political competition as an instrument for privatization, the paper finds that privatization improves firm productivity and profitability.
- **Economic takeaway.** Privatization is not just a financial or efficiency decision. It is a political allocation problem in which governments trade off dispersed economic gains against geographically concentrated political costs and patronage benefits.

2 Data and measurement (key objects)

2.1 Policy setting: Indian privatization after 1991

- India began major economic reforms in 1991 after a balance-of-payments crisis. Privatization, officially called "disinvestment," was part of this reform package.

- The paper focuses on firms owned by the federal government, which accounted for a large share of government-owned firm assets.
- In 1991, India had **280** nonfinancial firms owned by the federal government.
- Between fiscal years 1991 and 2006, **50** of these firms were privatized.
- The paper's main sample period runs from fiscal year 1990 to fiscal year 2004.
- The Congress Party government initiated the privatization program in 1991.
- The first major phase, 1991–1995, involved mostly **partial privatization**: minority equity sales without transfer of management control.
- The BJP government restarted privatization after 1999 and undertook more **control-transfer privatizations**: majority sales and transfer of management control to private owners.

Interpretation. The institutional setting lets the authors study both partial privatization and control transfer, but the main empirical idea is the same: the government chooses which firms to expose to private ownership and market monitoring.

2.2 Firm-level financial data

- The authors observe financial data for **259** of the **280** manufacturing and nonfinancial service-sector firms owned by the federal government.
- The data cover roughly **92%** of federal government-owned firms.
- The financial data come from company annual reports collected by the Centre for Monitoring the Indian Economy.
- The paper observes financial data for **49** of the **50** federal government-owned firms privatized between 1991 and 2004.
- The sample excludes firms located in Jammu and Kashmir because elections were not always held there during the sample period.
- For most years, firms must satisfy stock-market listing requirements: they need positive profits in at least 3 of the preceding 5 years. This restriction is relaxed for 1999–2003 because some privatizations were private asset sales rather than public listings.
- A privatized firm is included only until the year of its first sale to private owners. This avoids attributing postprivatization performance changes to preprivatization selection variables.

2.3 Political and electoral data

- The authors collect electoral data for each of India's **543** single-member parliamentary districts.

- They use federal elections held in 1991, 1996, 1998, 1999, and 2004.
- They hand-collect the address of each firm's main operations and match firms to electoral constituencies.
- About **80%** of the companies have their main operations in only one electoral constituency.
- For firms with multiple plants, the main plant is defined as the plant with the largest asset base.
- The paper uses digital geographic mapping to construct political measures around the firm's location.
- The main political variables are measured for all electoral constituencies within a **10 km radius** of the firm's main operations.
- Robustness checks use 0 km, 5 km, 25 km, 50 km, 100 km, and state-level political measures.

2.4 Key political variables

Let i index firms and $r(i)$ denote the geographic region around firm i's main operations.

- **Government vote share:**

$$GovtVote_{i,t} = \text{vote share of the governing party alliance in } r(i). \tag{1}$$

- **Opposition vote share:**

$$OppVote_{i,t} = \text{vote share of the main opposition party alliance in } r(i). \tag{2}$$

- **Vote-share difference:**

$$VoteDiff_{i,t} = GovtVote_{i,t} - OppVote_{i,t}. \tag{3}$$

  Lower values mean that the opposition is close to, or ahead of, the governing party. In the sample, the 25th percentile is close to zero, so low values mainly capture competitive races.

- **Absolute vote-share difference:**

$$AbsVoteDiff_{i,t} = |GovtVote_{i,t} - OppVote_{i,t}|. \tag{4}$$

  Lower values mean a closer race, regardless of which alliance is ahead.

- **Firm importance:**

$$FirmImportance_{i,t} = \frac{Sales_{i,t}}{\sum_{j \in r(i)} Sales^{SOE}_{j,t}}. \tag{5}$$

This captures how important a firm is relative to other government-owned firms in the local region.

Interpretation. The key political variable is local electoral competitiveness. The more pivotal the constituency, the more costly it may be for the ruling party to impose concentrated privatization losses there.

2.5 Descriptive facts

- Privatized firms are much larger than firms that remain fully government-owned.
- Average annual sales of privatized firms are about **28,862** million rupees versus **6,777** million rupees for fully government-owned firms.
- Average assets of privatized firms are about **51,545** million rupees versus **14,436** million rupees for fully government-owned firms.
- Privatized firms have lower wage bills relative to sales: about **0.140** versus **0.966** for fully government-owned firms.
- The average government vote share around firms is about **0.396**; the average opposition vote share is about **0.308**.
- The average vote-share difference is about **0.088**; the 25th percentile is about **-0.006**, meaning many firms are in areas where the governing and opposition alliances are close.

2.6 Unobservables and timing

- **Selection problem.** Firms are not randomly privatized. Better, larger, or more politically convenient firms may be chosen first.
- **Local political costs.** Job losses and opposition to privatization may be concentrated around the firm's location, but the economic benefits of privatization are spread nationally.
- **Patronage benefits.** Politicians may use government-owned firms to reward supporters through employment, procurement, or local influence.
- **Timing.** Financial variables are lagged 1 year in the main hazard regressions.
- **First sale as event.** The event time in the main analysis is the first sale of equity to private owners, not later repeated sales.
- **Political information.** Political variables are updated using the most recent federal election.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic framework: dispersed benefits and concentrated costs

The paper's political-economy logic can be summarized as follows.

- Privatization produces broad benefits: fiscal revenue, efficiency gains, capital-market development, and better monitoring.
- Privatization also creates local concentrated costs: layoffs, lower wage rents, opposition from organized labor, and the loss of political control over firms.
- A government that cares about elections may avoid privatizing firms in politically competitive areas because a small local backlash can flip a parliamentary seat.
- A politician who benefits from patronage may resist privatizing firms located in his or her home state.

The decision rule is not written as a formal structural model, but the implicit comparison is:

$$\text{Privatize firm } i \quad \Longleftrightarrow \quad B_i^{economic} - C_i^{political} - R_i^{patronage} > 0, \tag{6}$$

where:

- $B_i^{economic}$ includes sale proceeds and expected efficiency benefits;
- $C_i^{political}$ includes expected electoral losses from local backlash;
- $R_i^{patronage}$ is the rent from retaining political control over the firm.

Interpretation. Even if privatization raises aggregate surplus, the government may delay it when the local electoral penalty or patronage loss is large.

3.2 Main hazard model for the privatization decision

The baseline empirical model is a Cox proportional hazard model:

$$h_i(t) = h_0(t) \exp\left(\beta_1 X_{i,t-1} + \beta_2 P_{i,t} + \beta_3 D_{i,t} + \delta_{industry(i)}\right), \tag{7}$$

where:

- $h_i(t)$ is the hazard that firm i is privatized in year t;
- $h_0(t)$ is the unspecified baseline hazard;

- $X_{i,t-1}$ contains lagged firm financial variables;
- $P_{i,t}$ contains political variables around the firm's location;
- $D_{i,t}$ contains demographic and regional controls;
- $\delta_{industry(i)}$ are industry fixed effects.

The firm-level controls include:

- log sales;
- profit divided by sales;
- wages divided by sales.

The demographic and regional controls include:

- log state per capita income;
- state per capita income growth;
- literacy around the firm's location;
- urbanization around the firm's location;
- firm importance in the local region.

Interpretation. A positive coefficient means that the variable raises the privatization hazard, so the firm is privatized earlier. A negative coefficient means privatization is delayed.

3.3 Financial variables in the hazard model

A simplified financial-only version is:

$$h_i(t) = h_0(t) \exp\left(\beta_1 \log(Sales_{i,t-1}) + \beta_2 \frac{Profit_{i,t-1}}{Sales_{i,t-1}} + \beta_3 \frac{Wages_{i,t-1}}{Sales_{i,t-1}} + Controls\right). \tag{8}$$

- If larger firms are easier to sell or attract more investor demand, then $\beta_1 > 0$.
- If governments want higher sale proceeds and successful early privatizations, then more profitable firms may be privatized earlier.
- If workers oppose privatization and are better organized in high-wage-bill firms, then $\beta_3 < 0$.

Interpretation. This specification treats privatization partly like an IPO decision, but with an additional labor-resistance margin.

3.4 Political competition specifications

The key political specifications add one of several political variables:

$$h_i(t) = h_0(t) \exp\left(\cdots + \gamma_1 GovtVote_{i,t}\right), \tag{9}$$

$$h_i(t) = h_0(t) \exp\left(\cdots + \gamma_2 OppVote_{i,t}\right), \tag{10}$$

$$h_i(t) = h_0(t) \exp\left(\cdots + \gamma_3 VoteDiff_{i,t}\right), \tag{11}$$

or

$$h_i(t) = h_0(t) \exp\left(\cdots + \gamma_4 AbsVoteDiff_{i,t}\right). \tag{12}$$

Predicted signs:

- If governments avoid districts where the opposition is strong, then $\gamma_2 < 0$.
- If governments avoid close races, then $\gamma_3 > 0$ and $\gamma_4 > 0$ because larger vote differences mean less competition.
- The sign on $GovtVote$ is ambiguous: the government might reward supporters by delaying privatization, or it might feel safer privatizing where it has stronger support.

Interpretation. The political competition variables ask whether a government delays privatization when the electoral consequences of local backlash are most dangerous.

3.5 Patronage test

The patronage test uses a minister-firm match:

$$HomeStateMatch_{i,m} = \mathbf{1}\left\{\text{firm } i \text{ is located in minister } m\text{'s home state}\right\}. \tag{13}$$

The outcome is:

$$Privatized_{i,m} = \mathbf{1}\left\{\text{firm } i \text{ is privatized while under minister } m\text{'s jurisdiction}\right\}. \tag{14}$$

The empirical comparison is a two-way tabulation:

$$Corr(Privatized_{i,m}, HomeStateMatch_{i,m}). \tag{15}$$

Interpretation. If politicians use government firms for patronage in their home states, then the correlation should be negative. The striking fact is stronger: no home-state matched firm is privatized.

3.6 Election-year and robustness specifications

To test whether elections directly slow privatization, the paper estimates an exponential hazard model with an election-year indicator:

$$h_i(t) = \exp\left(\alpha + \lambda ElectionYear_t + \theta VoteDiff_{i,t} + Controls\right). \tag{16}$$

The paper also tests whether the main political result survives controls for:

- Communist party vote share;
- whether the federal governing party also controls the state assembly;
- different geographic radii around the firm;
- control-transfer privatizations only;
- nonlinear firm-size controls;
- winsorized firm variables;
- restricting the sample to large Indian states.

Interpretation. These tests ask whether the political competition result is really about local electoral incentives rather than omitted region characteristics, left-party ideology, or the method of privatization.

3.7 Instrumental-variable performance regression

Because privatization is endogenous to firm characteristics, the paper uses political competition as an instrument for privatization in a treatment-effects/IV framework.

The first stage is:

$$Privatized_i = \pi_0 + \pi_1 VoteDiff_i + \pi_2 X_i + \pi_3 D_i + \eta_i. \tag{17}$$

The second stage is:

$$\Delta Performance_i = \alpha + \tau \widehat{Privatized}_i + X_i'\beta + D_i'\rho + \varepsilon_i. \tag{18}$$

Performance outcomes include:

- change in profits/assets;
- change in profits/sales;
- change in profits/wages.

Interpretation. The IV strategy uses local political competitiveness to isolate variation in privatization that is not simply driven by firms being larger, more profitable, or easier to sell.

4 Identification and instruments (what makes the estimates credible)

4.1 What is hard to identify

- Privatization is not randomly assigned.
- More attractive firms may be privatized earlier because they generate higher sale proceeds.
- Less politically costly firms may be privatized earlier because they face weaker worker opposition.
- Regions differ in income, education, urbanization, ideology, union strength, and growth opportunities.
- Politicians may choose privatization timing based on unobserved local factors that are correlated with electoral competitiveness.

Core empirical challenge. A raw comparison of privatized and nonprivatized firms confounds the effect of privatization with selection by financial and political characteristics.

4.2 What identifies the political competition effect

The paper's identification comes from matching firm locations to constituency-level electoral outcomes and asking whether firms in politically competitive regions are privatized later, conditional on firm and regional characteristics.

Key controls include:

- lagged firm sales, profitability, and wages/sales;
- industry fixed effects;
- the Cox baseline hazard, which absorbs common time effects;

- regional income, income growth, literacy, and urbanization;
- firm importance to the local region.

Interpretation. The identifying comparison is between similar government-owned firms in similar broad conditions, but located in areas with different degrees of local electoral competition.

4.3 Why vote-share difference is informative

- In a winner-take-all parliamentary district, small swings in voter support can change which party wins the seat.
- Privatization may generate a local backlash through worker opposition or public opposition to selling state assets.
- This backlash matters most where the governing and opposition alliances are close.
- Therefore, if politics matters, the government should delay privatization in low-$VoteDiff$ or low-$AbsVoteDiff$ areas.

Interpretation. Political competition is a proxy for the marginal electoral cost of privatizing a local firm.

4.4 What identifies the patronage result

The patronage evidence uses the identity of the cabinet minister with jurisdiction over each firm.

- If a firm's main operations are located in the minister's home state, the minister may obtain greater private political benefits from keeping the firm government-owned.
- The paper compares privatization outcomes for minister-firm pairs with and without a home-state match.
- The fact that no home-state matched firm is privatized is strong evidence that political control itself matters.

Interpretation. The patronage test is not just about party vote shares. It directly connects the politician controlling the firm to the geographic location of the firm.

4.5 Instrument validity for the privatization-performance effect

The IV performance result requires two assumptions:

- **Relevance.** Vote-share difference must predict privatization. This is supported by the first stage: higher vote-share difference raises the probability of privatization.

- **Exclusion.** Conditional on controls, vote-share difference should affect firm performance only through privatization, not through other channels.

Why the exclusion restriction is plausible:

- The political variable is local constituency-level electoral competition, while firm performance is a firm-level outcome.
- The regressions control for firm financial characteristics, demographics, industry effects, and year effects.
- The political variable is used to capture government selection, not market demand or operating productivity directly.

Main limitation:

- Local political competition could still be correlated with unobserved local labor pressure, local public-sector dependence, or political bargaining that affects firms independently of privatization.

4.6 Limitations and interpretation

- The main privatization results are drawn largely from the first privatizing government, because most sales occurred in the 1991–1995 Congress period.
- The paper studies Indian federal government-owned firms, so external validity depends on whether other countries have similar electoral and patronage incentives.
- The IV performance analysis has a smaller sample than the hazard analysis.
- Political variables are measured geographically around the firm's main operations; political effects could be broader if firms have workers, suppliers, or stakeholders in many regions.
- The paper cannot fully observe informal political bargaining, union pressure, or ministerial intervention.

4.7 Relation to the literature

- **Privatization literature.** The paper complements work showing that privatization improves firm performance by studying which firms are selected for privatization.
- **IPO literature.** The financial variables parallel the private-firm IPO literature: larger firms may be easier to sell because information asymmetry is lower.
- **Political economy of finance.** The paper contributes to work showing that financial-market reforms are shaped by electoral incentives, interest groups, and political rents.

- **Patronage and state ownership.** The paper provides direct evidence that politician-firm links affect the decision to privatize.

5 Tables and figures

5.1 Figure 1: electoral district map of India

Read:

- Figure 1 shows the geographic structure of India's parliamentary constituencies.
- The point of the figure is methodological: the paper's identification uses within-country variation across many local electoral districts.
- Because India has hundreds of constituencies and substantial regional political variation, firm location can be linked to local political competition.

Figure 1. Electoral district map of India.

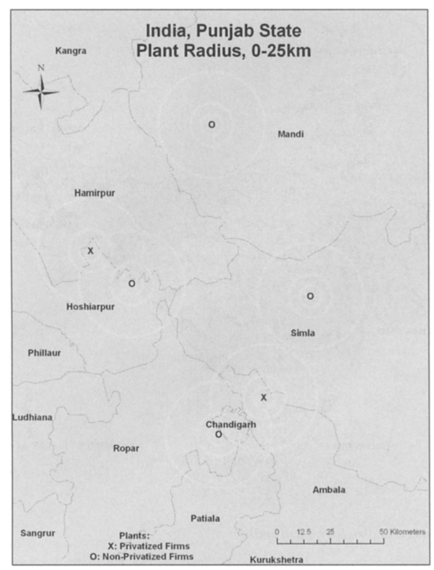

Figure 2. Electoral constituencies at the firm level. The figure illustrates how the political variables are constructed at varying distances around the location of firms in one state.

5.2 Figure 2: constructing electoral constituencies around firms

Read:

- Figure 2 illustrates how the authors construct political variables around the main operations of a firm.
- The example uses firms in Punjab and shows constituencies at different distances from firm locations.

- This matters because political costs may not be confined to the exact constituency where the plant sits. Workers, suppliers, and voters affected by the firm may live nearby.
- The main analysis uses a 10 km radius, and Table V checks other radii.

5.3 Figure 3: privatization frequency and political competition

Read:

- Figure 3 plots the frequency of privatization against $VoteDiff$ for 1991–1995.
- The visual relationship is positive: privatization is less common when the governing and opposition party alliances are in a close race.
- This figure previews the main regression result in Table III.
- The figure is not the final evidence because it does not fully control for firm and regional characteristics, but it makes the political mechanism easy to see.

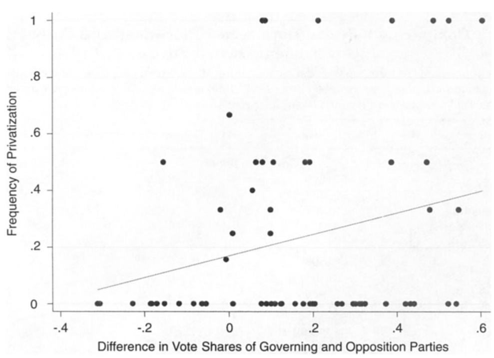

Figure 3. Relationship between frequency of privatization at the electoral constituency level and political variables. The graph depicts the relationship between the frequency of privatization in electoral constituencies within 10 km of the main operations of all firms, constructed as the number of privatized firms/number of government-owned and privatized firms in the region. The data correspond to fiscal years 1991 to 1995. Industries in which no firms are privatized and firms that do not meet the listing requirement are excluded from the analysis.

Table I

Comparing Privatized and Fully Government-Owned Firms

This table presents sample statistics of the firm-specific financial variables used in the analysis for fiscal years 1990 to 2004, where fiscal year 2004 runs from April 2004 to March 2005. All variables are described in Appendix Table A. Firms that do not meet the listing requirement for all the years except 1999 to 2003 are excluded from the analysis and only firm-years for which we have sales, profits, and wages data are included. Standard deviations are in parentheses.

Variables	Privatized	Fully Government-Owned	All Firms
	Sales		
Mean	28,862***	6,777	8,910
SD	(51,935)	(23,652)	(28,407)
Number of firm-years	153	1,431	1,584
	Assets		
Mean	51,545***	14,436	18,025
SD	(114,022)	(55,067)	(64,103)
Number of firm-years	153	1,429	1,582
	Profit/sales (%)		
Mean	0.184	1.301	1.193
SD	(.189)	(59.777)	(56.816)
Number of firm-years	153	1,431	1,584
	Wages/Sales (%)		
Mean	0.140***	0.966	0.887
SD	(.162)	(9.176)	(8.725)
Number of firm-years	153	1,431	1,584
Number of firms	49	191	240

***Indicates significance at the 1% level.

5.4 Table I: comparing privatized and fully government-owned firms

Read:

- Privatized firms are substantially larger than firms that remain fully government-owned.
- Average sales are about **28,862** million rupees for privatized firms versus **6,777** million rupees for fully government-owned firms.
- Average assets are about **51,545** million rupees for privatized firms versus **14,436** million rupees for fully government-owned firms.
- Privatized firms have lower wages/sales, about **0.140** versus **0.966**.

- Profit/sales is not clearly different in the raw comparison.
- The table shows why selection matters: the firms chosen for privatization look different before privatization.

5.5 Table II: comparing political data around firms

Read:

- The average government vote share around firms is about **0.396**; the average opposition vote share is about **0.308**.
- Privatized firms tend to be in areas with somewhat lower opposition vote share.
- The average vote-share difference is positive, but the 25th percentile is close to zero.
- This means that a sizable part of the sample involves politically competitive areas.
- The table motivates using both $VoteDiff$ and $AbsVoteDiff$ as political competition measures.

Table II

Comparing Political Data across Privatized and Fully Government-Owned Firms

This table presents the summary statistics describing the political variables for the five federal elections held in the fiscal years 1990, 1995, 1997, 1998, and 2003. All variables are described in Appendix Table A. Standard deviations are in parentheses.

Variables	All Firms	Fully Government-Owned	Privatized
	Govt Vote Share		
Mean	0.396	0.395	0.399
SD	(0.148)	(0.149)	(0.143)
75^{th} percentile	0.487	0.489	0.470
25^{th} percentile	0.372	0.362	0.396
	Opposition Vote Share		
Mean	0.308	0.310	0.291
SD	(0.151)	(0.148)	(0.172)
75^{th} percentile	0.420	0.420	0.402
25^{th} percentile	0.157	0.173	0.105
	Vote Share Difference		
Mean	0.088	0.085	0.108
SD	(0.228)	(0.226)	(0.238)
75^{th} percentile	0.256	0.256	0.301
25^{th} percentile	−0.006	−0.011	−0.006
	Abs Vote Share Difference		
Mean	0.191	0.191	0.191
SD	(0.151)	(0.148)	(0.177)
75^{th} percentile	0.314	0.312	0.329
25^{th} percentile	0.139	0.142	0.099
Number of firm-years	1,579	1,426	153

Table III

The Decision to Privatize: The Role of Financial and Political Factors

This table presents results from estimating a Cox proportional hazard regression covering fiscal years 1990 to 2004. The firm-specific variables, described in Appendix Table A, are lagged 1 year. We use natural logs of the firm-specific financial variables. Firms that do not meet the listing requirement except from 1999 to 2003 are excluded. Heteroskedasticity-robust standard errors, clustered at the state level, are in parentheses.

	(1)	(2)	(3)	(4)	(5)
Ln(sales)	0.630***	0.630***	0.721***	0.686***	0.686***
	(0.153)	(0.147)	(0.134)	(0.138)	(0.143)
Profit/sales	0.024	0.024	0.019	0.02	0.022
	(0.038)	(0.037)	(0.039)	(0.041)	(0.037)
Wages/sales	−3.336*	−3.200*	−3.169*	−3.002*	−3.233*
	(1.965)	(1.744)	(1.755)	(1.672)	(1.905)
ln(per capita income)	0.304	0.296	0.342	0.312	0.375
	(0.499)	(0.485)	(0.565)	(0.534)	(0.547)
Per capita income	−0.261	−0.269	−0.127	−0.199	−0.183
Growth	(0.517)	(0.540)	(0.628)	(0.600)	(0.620)
Literacy	3.039*	3.052**	2.669***	2.787***	2.718**
	(1.590)	(1.494)	(0.995)	(1.067)	(1.122)
Urbanization	−1.866***	−1.931***	−1.351**	−1.669***	−1.664***
	(0.715)	(0.621)	(0.639)	(0.582)	(0.627)
Firm importance	−0.402	−0.444	−0.405	−0.495	−0.464
	(0.480)	(0.393)	(0.401)	(0.374)	(0.374)
Govt vote share		0.967			
		(1.386)			
Opposition vote share			−2.914***		
			(0.694)		
Vote share difference				1.789***	
				(0.656)	
Abs vote share difference					1.680***
					(0.554)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Number of firms	239	239	239	239	239
Number of firm-years	1,579	1,579	1,579	1,579	1,579

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

5.6 Table III: main hazard regressions

Read:

- This is the paper's main table on the decision to privatize.
- Log sales has a positive and significant coefficient in all specifications. Larger firms are privatized earlier.

- Wages/sales has a negative and significant coefficient. Firms with larger wage bills are privatized later.
- Profit/sales is not statistically significant in the main specifications.
- Government vote share is positive but not significant.
- Opposition vote share has a negative and significant coefficient, about **-2.914**. Privatization is delayed where opposition support is higher.
- $VoteDiff$ has a positive and significant coefficient, about **1.789**. Privatization is faster where the governing party is less electorally threatened.
- $AbsVoteDiff$ also has a positive and significant coefficient, about **1.680**. Privatization is slower in close races regardless of which alliance is ahead.
- Regional controls matter too: privatization is faster in more literate areas and slower in more urbanized areas.

Economic magnitude:

- The paper reports that the privatization rate is roughly **1.5 times higher** for firms located at the 75th percentile of political competition relative to the 25th percentile, where lower percentiles correspond to closer races.
- For opposition vote share, moving from the 25th percentile to the 75th percentile reduces the privatization rate to less than half.

5.7 Table IV: political patronage

Read:

- Table IV directly tests whether a firm is privatized when its main operations are in the home state of the minister responsible for that firm.
- During the Congress period, there are **15** cases where the firm's state matches the minister's home state; **zero** are privatized.
- Pooling Congress and BJP periods, there are **45** home-state match cases; again, **zero** are privatized.
- The correlation between home-state match and privatization is negative and statistically significant.
- This is the clearest evidence in the paper that politicians value control over government-owned firms for patronage purposes.

Table IV

The Decision to Privatize: The Role of Political Patronage

This table presents a two-way tabulation and correlation analysis between the decision to privatize a firm and the home states of the ministers who have jurisdiction over that firm. It excludes the industries in which no privatizations occur. Each minister–firm pair is taken as a single observation regardless of the length of time the firm remains under that minister's jurisdiction. *Main Operations in Home State* is a dummy variable equal to one if the state where the firm's main operations are located is the same as the state from which the Cabinet minister who has jurisdiction over that firm is elected. *Privatized* is a dummy variable that is equal to one if the firm is privatized while under the jurisdiction of a given minister. Once a firm is privatized it is dropped from the sample. Firms that do not meet the listing requirement for all the years except 1999 to 2003 are excluded from the analysis.

	Privatized		
Panel A: Congress Government (1991–1995)			
Main operations in home state	No	Yes	Total
No	133	38	171
Yes	15	0	15
Total	148	38	186
Correlation	−0.150**		
Panel B: Congress and BJP Governments (1991–1995 & 1999–2002)			
Main operations in home state	No	Yes	Total
No	518	46	564
Yes	45	0	45
Total	563	46	609
Correlation	−0.081**		

**Denotes statistical significance at the 5% level.

Table V

Political Measures at Different Distances from the Main Operations of the Firm

This table presents results from estimating a Cox proportional hazard regression covering fiscal years 1990 to 2004, where the political variables are measured for all electoral constituencies within different radii around the main operations of the firm. The political variable in the last regression is constructed using all the electoral constituencies in the state where the main operations of the firm are located, regardless of the distance. All the regressions include firm-specific controls (*Ln(Sales), Profit / Sales*, *Wages / Sales*) and demographic controls (*Ln(Per Capita Income)*, *Per Capita Income Growth*, *Literacy*, and *Urbanization* at different distances). The variables are defined in Appendix Table A. Firms that do not meet the listing requirement for all the years except 1999 to 2003 are excluded from the analysis. Heteroskedasticity-robust standard errors, clustered at the state level, are in parentheses.

Maximum Distance between the Electoral Constituencies and the Firm	0 Km	5 Km	25 Km	50 Km	100 Km	State
Vote share difference	0.964*	1.528***	1.843***	1.535**	1.511**	1.540**
	(0.547)	(0.580)	(0.705)	(0.699)	(0.742)	(0.686)
Firm-specific and demographic controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of firm-years	1,579	1,579	1,579	1,579	1,579	1,579
Number of firms	239	239	239	239	239	239

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

5.8 Table V: political variables at different distances

Read:

- Table V checks whether the political competition result depends on the choice of a 10 km radius.
- The authors recompute $VoteDiff$ using 0 km, 5 km, 25 km, 50 km, 100 km, and the state level.
- The coefficient on $VoteDiff$ remains positive and statistically significant across all these geographic definitions.
- This robustness check is important because the relevant political constituency for a firm may be broader than the exact electoral district where the firm is located.

5.9 Table VI: election years, Communist parties, and regional politics

Read:

- Column 1 adds an election-year dummy using an exponential hazard model.
- Privatization is significantly delayed in election years.
- The $VoteDiff$ result remains significant, showing that local political competition matters even apart from national election timing.
- Column 2 adds Communist party vote share. This variable is not significant, and $VoteDiff$ remains significant.
- Column 3 adds whether the federal governing alliance controls the state assembly. This variable is not significant.

- The table strengthens the interpretation that the main result is about electoral competition, not simply left-party strength or state-government alignment.

Table VI

Election Years, Communist Parties, and Regional Politics

This table presents results from estimating exponential and Cox proportional hazard regressions of the government's decision to privatize, covering fiscal years 1990 to 2004. All variables are described in Appendix Table A. Firms that do not meet the listing requirement for all the years except 1999 to 2003 are excluded from the analysis. The first regression also includes government fixed effects. Heteroskedasticity-robust standard errors, clustered at the state level, are in parentheses.

	(1)	(2)	(3)
Ln(sales)	0.702***	0.693***	0.685***
	(0.126)	(0.137)	(0.138)
Profit/sales	0.018	0.016	0.02
	(0.046)	(0.040)	(0.041)
Wages/sales	−3.126*	−3.049*	−2.987*
	(1.693)	(1.689)	(1.666)
Election year	−1.801*		
	(1.038)		
Communist vote share		0.87	
		(1.065)	
State assembly majority			−0.292
			(0.718)
Vote share difference	1.813**	1.700**	1.756***
	(0.705)	(0.705)	(0.666)
Demographics controls	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Number of firms	239	239	239
Number of firm-years	1,579	1,579	1,579
Hazard model	Exponential	Cox	Cox

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table VII

Privatization with the Transfer of Management Control (1999–2003)

This table presents the results from estimating a Cox proportional hazard regression of the government's decision to sell majority stakes and transfer management control in 17 firms starting in 1999. The political variables are constructed using data from the 1999 federal elections and the variables are described in Appendix Table A. Heteroskedasticity-robust standard errors, clustered at the state level, are in parentheses.

	(1)	(2)
Ln(sales)	0.169	0.167
	(0.197)	(0.203)
Profit/sales	−0.309	−0.280
	(0.251)	(0.238)
Wages/sales	−3.767	−4.014*
	(2.390)	(2.180)
Ln(per capita income)	0.202	0.291
	(0.442)	(0.531)
Per capita income growth	−1.329*	−1.371*
	(0.734)	(0.727)
Literacy	3.557*	5.034*
	(2.025)	(2.982)
Urbanization	−2.663	−3.584**
	(1.736)	(1.391)
Firm importance	−0.356	−0.490
	(0.893)	(0.915)
Vote share difference		2.422*
		(1.305)
Industry fixed effects	Yes	Yes
Number of firms	223	223
Number of firm-years	878	878

* and ** denote statistical significance at the 10% and 5% levels, respectively.

5.10 Table VII: privatization with transfer of management control

Read:

- Table VII focuses only on control-transfer privatizations, mostly during the BJP period from 1999 to 2003.
- These are stronger privatizations than minority stake sales because private owners receive management control.
- $VoteDiff$ remains positive and significant.
- The result suggests that electoral competition affects not only partial privatization, but also deeper privatization involving control transfer.
- Wages/sales remains negative, again showing that labor-related costs slow privatization.

5.11 Table VIII: impact of privatization on performance

Read:

- Table VIII uses $VoteDiff$ as an instrument for privatization.
- The first stage shows that $VoteDiff$ strongly predicts privatization. The first-stage coefficient is about **2.755** and statistically significant.
- In the second stage, privatization raises changes in profits/assets, profits/sales, and profits/wages.

Table VIII

Effect of Privatization on Firm Performance

This table presents the results from a pooled instrumental variable regression to analyze the effect of privatization on firm performance. Panel A reports results from the second-stage 2SLS regressions with firm performance measures as the dependent variables. The variable *Privatized*, equal to one if the firm is privatized by the government in power, is instrumented by the fitted probabilities obtained from a probit model, which has *Privatized* as the dependent variable and is reported in Panel B. The variable included in the probit model but excluded in the performance estimates is *Vote Share Difference*, described in Appendix Table A. The performance change variables measure the change in firm-level characteristics from the most recent year before the government is elected to the last year it remains in power. The dependent variables are winsorized 2.5% at each tail. If a firm is privatized by a previous government, it is not included for performance evaluation in the subsequent period. Firms that do not meet the listing requirement for all the years except 1999 to 2003 are excluded from the analysis. The remaining variables are as described in Appendix Table A. Standard errors are robust to clustering at the state level and all specifications include industry and year fixed effects.

Panel A: Second-Stage Regressions with Firm Performance			
Dependent Variable	Change in Profits/Assets	Change in Profits/Sales	Change in Profits/Wages
Privatized	0.415**	0.402**	2.480**
	(0.170)	(0.183)	(0.981)
Ln(sales)	−0.088***	−0.110**	−0.435
	(0.034)	(0.046)	(0.285)
Profit/sales	−0.077**	0.107***	0.290***
	(0.038)	(0.021)	(0.103)
Wages/sales	0.255**	0.479***	1.216***
	(0.102)	(0.090)	(0.335)
Constant	1.420*	1.654**	7.005**
	(0.751)	(0.769)	(3.349)
Demographic controls	Yes	Yes	Yes
Industry and year fixed effects	Yes	Yes	Yes
Number of firm-years	203	200	203

Panel B: First-Stage Probit	
Dependent Variable	Privatized
Vote share difference	2.755***
	(0.596)
Ln(sales)	0.885***
	(0.159)
Profit/sales	0.063
	(0.194)
Wages/sales	−1.277
	(1.990)
Constant	−11.410**
	(4.946)
Demographic controls	Yes
Industry and year fixed effects	Yes

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

- The estimated coefficient on privatization is about **0.415** for profits/assets, **0.402** for profits/sales, and **2.480** for profits/wages.
- The table addresses a key selection concern: if better firms are more likely to be privatized, simple before-after or treated-control comparisons may overstate privatization effects.
- The IV result says that, after correcting for endogenous selection using political variation, privatization still improves firm performance.

6 Short summary

- **Method.** The paper estimates Cox hazard models of the timing of privatization and uses firm location matched to electoral constituencies to measure local political incentives.
- **Data.** The setting is Indian federal government-owned firms from 1990 to 2004, combined with electoral data from 543 parliamentary districts and hand-collected firm-location data.
- **Financial selection.** Larger firms are privatized earlier; firms with higher wages/sales are privatized later.
- **Political selection.** Privatization is delayed where the opposition is stronger and where the governing and opposition alliances are in a close race.

- **Patronage.** Firms located in the home state of the minister in charge are never privatized.
- **Performance.** Using political competition as an instrument, privatization is found to improve profitability and efficiency.

7 Conclusion

7.1 Core conclusion

- The paper shows that governments do not privatize firms randomly.
- Financial characteristics matter: bigger firms are easier or more attractive to sell, while firms with large wage bills are politically harder to sell.
- Local electoral incentives matter: governments delay privatization in politically competitive constituencies.
- Patronage matters: politicians appear to protect firms located in their home states from privatization.
- Correcting for this selection with political instruments, the paper finds that privatization improves firm performance.

7.2 Economic intuition

- Privatization creates broad economic gains but concentrated political losses.
- In close electoral races, even a small local backlash can be costly to the governing party.
- Workers in state-owned firms are concentrated, organized, and visible, while taxpayers and consumers who benefit from privatization are dispersed.
- Politicians also lose control rights over firms after privatization, which weakens patronage channels.
- Therefore, privatization timing reflects both efficiency considerations and political incentives.

7.3 Takeaway

This paper connects corporate finance, privatization, and political economy. It shows that the decision to privatize is shaped not only by firm fundamentals, but also by electoral competition and political control. For studying privatization effects, this matters because treated firms are selected. For policy, it means that delays in privatization may persist even when privatization is efficient, because the political costs are local and concentrated while the benefits are broad and diffuse.